Duration : 3 Hours **Marks : 80 Marks**

N.B. : (1) Question No 1 is Compulsory.
(2) Attempt any three questions out of the remaining five.
(3) All questions carry equal marks.
(4) Assume suitable data, if required and state it clearly.

Q.1 Solve any four.

	a.	Compare and contrast Boolean Model vs Vector Space Model.	**5**
	b.	Specify the significance of User Relevance feedback in an IR system.	**5**
	c.	Explain inverted file indexing with suitable examples.	**5**
	d.	Explain the process of Structured Text retrieval model.	**5**
	e.	Illustrate different types of keyword-based queries.	**5**
Q.2	a.	Draw the taxonomy of IR models and explain any one IR modeling technique.	**10**
	b.	What is the significance of **tf** and **idf** ? How can you calculate **tf** and **idf** in a vector model?	**10**
Q.3	a.	Explain the various system related issues faced in Information retrieval systems and how they can be refined for a deployed system.	**10**
	b.	State the different types of queries. Explain the pattern matching query concept with an example .	**10**
Q.4	a.	What is the role of suffix array and suffix tree in information retrieval system with example.	**10**
	b.	What is Latent Semantic Indexing model ? Write the advantages of Latent Semantic Indexing Model?	**10**
Q.5	a.	Define Multimedia information retrieval. Discuss indexing and searching.	**10**
	b.	What is the difference between Unranked Retrieval models and Ranked Retrieval models.	
Q.6		Write short notes on any two.	**20**
	a.	Information Retrieval in digital libraries.	
	b.	Sequential Searching.	
	c.	Flat browsing vs Hypertext Browsing model.	
	d.	Distributed Information Retrieval.	

COMP/VII | R-19 12/6/24

Paper / Subject Code: 42178 / INFORMATION RETRIEVAL (DLOC - IV)

Duration: - 3 Hours **Marks: 80 Marks**

NB: - Question 1 is compulsory
Solve any four questions from Question no. 1.
Solve any three questions from the remaining.

1	a.	Describe the differences between information and data in the context of Information Retrieval.	20 (4x5)
	b.	Explain the process of Structured text retrieval model.	
	c.	List the key components of the vector space model.	
	d.	What are different types of queries in Information Retrieval?	
	e.	Explain multimedia indexing approach?	
2	a.	What is Information Retrieval Model? Explain the taxonomy of Information Retrieval Models	10
	b.	Discuss the relative advantages and disadvantages of the Classic Information Retrieval compared to the Alternative Probabilistic Models. Is the probabilistic model always superior to the Classic Information Retrieval?	10
3	a.	Explain in detail how information retrieval systems utilize user feedback to enhance the search results.	10
	b.	Explain the role of a suffix tree in the indexing of Information Retrieval, Pattern matching, Structured queries with example	10
4	a.	Explain the function and advantages of using parametric and zone indexes in enhancing information retrieval.	10
	b.	What is the significance tf-idf weight? Can the tf-idf weight of a term in a document exceed 1? Why?	10
5	a.	Compare and contrast evaluation of ranked and unranked Retrieval Results?	10
	b.	Explain the process of query processing in distributed information retrieval systems.	10
6		Write short notes on any two	20
	a.	Inexact top K document retrieval	
	b.	Automatic local analysis vs Automatic global analysis	
	c.	Latent Semantic Indexing Model	
	d.	Flat browsing vs hypertext browsing model.	

Paper / Subject Code: 42178 / INFORMATION RETRIEVAL (DLOC - IV)

R-19 - VII - Comp - 10-06-25

Duration: - 3 Hours | Marks: 80 Marks

NB - Question 1 is compulsory
Solve any four questions from Question no. 1.
Solve any three questions from the remaining.

1	a.	Explain the difference between information and data with suitable examples. How does this distinction influence the design of IR systems?	5
	b.	Explain the retrieval process in digital libraries and how it differs from traditional libraries.	5
	c.	Identify and explain the main advantages of the vector space model over Boolean models in IR	5
	d.	What are different types of queries in Information Retrieval?	5
	e.	Differentiate between Multimedia IR and Text-based IR.	5
2	a.	What are the various categories of information retrieval models? Provide a detailed explanation of probabilistic models.	10
	b.	Compare the advantages of Algebraic models over Classical IR models. Can algebraic models be more suitable in certain contexts?	10
3	a.	Explain the process of user relevance feedback and how it improves retrieval performance. Illustrate with a sample interaction loop.	10
	b.	Explain the role of a suffix tree in the indexing of Information Retrieval, Pattern matching, Structured queries with example	10
4	a.	Compare and contrast the use of parametric indexes and zone indexes in optimizing retrieval in structured text databases. In what situations would one be more beneficial than the other?	10
	b.	Explain how Tf-idf weighting helps in ranking documents. Include formula and example.	10
5	a.	How do precision and recall metrics differ when evaluating the effectiveness of ranked and unranked retrieval results?	10
	b.	Explain the architecture of a Distributed Information Retrieval system. What are its advantages?	
6		Write short notes on any two	20
	a.	Inexact top K document retrieval	
	b.	Automatic local analysis vs Automatic global analysis	
	c.	The strengths and weaknesses of Probabilistic Latent Semantic Indexing	
	d.	Flat browsing vs hypertext browsing model.	

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Paper / Subject Code: 42178 / INFORMATION RETRIEVAL (DLOC - IV) 21/11/24

COMP / VII / R-19

Duration: - 3 Hours **Marks: 80 Marks**

NB: - Question 1 is compulsory

Solve any four questions from Question no. 1.

Solve any three questions from the remaining.

1 **a.** Define information retrieval and elaborate on the taxonomy of Information retrieval Model. **20**

b. Explain multimedia indexing approach.

c. Explain the process of Structured Text retrieval model.

d. Specify the significance of User Relevance feedback in an IR system.

e. Specify the need for pattern matching in Information retrieval.

a. Draw the taxonomy of IR models and explain any one IR modeling technique. **10**

b. Explain query processing in context of distributed IR **10**

a. Compare and contrast between automatic local analysis and global analysis? Elaborat on any one with an example **10**

b. Define what is meant by an Inverted file with an example. Explain it with a neat Diagram. **10**

a. Elaborate on how the ranked and unranked retrieval sets are evaluated. . **10**

b. Explain the method for calculating tf and idf in a vector model. Further elaborate the need for tf and idf in an information system **10**

a. With neat diagrams, elaborate on the role of suffix array and suffix tree in any given information retrieval system. **10**

b. Specify what is meant by Latent Semantic Indexing Model. Further , mention the advantages of Latent Semantic Indexing Model **10**

Write short notes on **any two** **20**

a. Significance tf-idf weight

b. Flat browsing vs Hypertext Browsing model.

c. Parametric and zone indices

d. Roccio method for query expansion

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Paper / Subject Code: 42178 / INFORMATION RETRIEVAL (DLOC - IV)

Duration: - 3 Hours **Marks: 80 Marks**

NB: - Question 1 is compulsory
Solve any four questions from Question no. 1 .
Solve any three questions from the remaining.

1	a.	Define information retrieval and list down classification of information retrieval systems?	20 (4x5)
	b.	Explain the process of Structured Text retrieval model.	
	c.	Explain the taxonomy of Information retrieval Model.	
	d.	Explain the role of pattern matching in Information retrieval.	
	e.	Explain multimedia indexing approach ?	
2	a.	Illustrate information retrieval system? Discuss its relationship to DBMS, digital libraries and data warehouses?	10
	b.	Explain in detail about vector-space retrieval models with an example?Is the vector space model always superior to the Boolean model?	10
3	a.	What is local and global analysis and Differentiate between automatic local analysis and global analysis?	10
	b.	What is the role of suffix array and suffix tree in information retrieval system with example.	10
4	a.	What is the inverted file? Explain in detail with example.	10
	b.	What is the significance tf-idf weight? Can the tf-idf weight of a term in a document exceed 1? Why?	10
5	a.	Compare and contrast evaluation of ranked and unranked Retrieval Results ?	10
	b.	Define multimedia information retrieval. Discuss indexing and searching?	10
6		Write short notes on any two	20
	a.	Inexact top K document retrieval	
	b.	Parametric and zone indices	
	c.	Latent Semantic Indexing Model	
	d.	Flat browsing vs hypertext browsing model.	
